Francisco José Díaz Villarreal

Software Engineer | Full Stack Developer | Spec Driven Development | Project Lead | Data & Game Specialist
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Software Engineer, Spec Driven and **Full Stack Developer** with **PhD in Applied Sciences** and solid experience in the design and development of **scalable web applications**, **RESTful APIs** and systems based on data science and video games (chess-based systems). Specialized in **Spec Driven** and **Full Stack Development**, system integration and application of good software engineering practices. Experience working with multiple languages, frameworks, applying **Github** and modern software architecture. Profile with strong analytical skills, technical leadership and experience guiding teams in development environments with a focus on real products.

➢ PROFESSIONAL EXPERIENCE

✓ **Full Stack Developer & Software Engineering Instructor**
Universidad Tecnológica Nacional (UTN) – CUDI | 2023 – 2025
• Design and development of Full Stack web applications using **Vue.js, React.js, frontend Bootstrap.js** and **Node.js, C, C++, C#** in backend
• Building and integrating **RESTful APIs** connected to **SQL and NoSQL databases**
• Applying workflows with **Github**
• Database design **implementation**, system integration and application structure

✓ **SQL & Database Developer (Academic Projects)**
Universidad Argentina de la Empresa (UADE) | 2021 – 2022
• Design and optimization of **relational database schemas** to ensure information integrity
• Developing and optimizing **SQL Server Management Studio** to improve data performance and efficiency
• Application of **data normalization** principles and good relational modeling practices

✓ **Systems & Software Support Engineer**
Servicio Autónomo de Registros y Notarías (SAREN) | 2009 – 2019
• Support and maintenance of enterprise information systems used in administrative processes
• Improvement of digital document management processes, contributing to information management
• Implementation of technical solutions to optimize system reliability and operational continuity

✓ **Project Lead – Masterdrez Science Video Game**
Centro Nacional de Desarrollo e Investigación en Tecnología (CENDITEL) | 2008 - 2012
• Coordinator responsible for and technical management of the development of the video game Masterdrez
• Concept Author, Software Architecture Definition, Game System Logic and Design
• Supervision , implementation, technical direction of a team of engineers with master's degree

✓ **Software Development Instructor & Technical Mentor**
Universidad Bolivariana de Venezuela (UBV) | 2004 – 2010
• Training in programming, software engineering and computer architecture
• Mentoring teams in software projects , game development, promoting development
• Technical coordination of applied development projects aimed at building solutions

✓ **Computer Science & Chess Systems Instructor | Strategic Thinking & Game Logic**
Unidad Educativa "Canónigo Uzcátegui" | 2004 - 2009
• Taught computer fundamentals, logic, and basic programming structures
• I applied chess concepts as a model of strategic systems and decision-making
• Coordinated educational initiatives at the municipal level related to health and cognitive development
• Led the implementation of the National School Chess Program

➢ ACADEMIC BACKGROUND

- ✓ **Universidad de los Andes (ULA)**
  PhD: Doctor of Applied Science – Degree – (2024)
- ✓ **Instituto Superior Politécnico José Antonio Echeverría (ISPJAE)**
  M. Sc.: Master's Degree in Applied Computer Science – Degree – *(2010)*
- ✓ **Universidad de Buenos Aires (UBA)**
  Spe.: Diploma in Initialization in Programming and Data Analysis (Python) – Degree – (2025)
- ✓ **Universidad Valle del Momboy (UVM)**
  Eng.: Computer Engineering – Degree – *(2004)*
- ✓ **Centro de Formación Profesional (CFP)** - Vocational Technical Education Level: 3
  Spe.: Professional Specialization in Video Game Programming (C#) – Certificate – (2022)
  Spe.: Programmer Track (TypeScript and SQL) – Certificate – (2021)

➢ COMPUTER COURSES

- **Adaptive Behavior** – Certificate
- **Manual, Agile, and Tooled Testing** – Certified

➢ COMPUTER CERTIFICATIONS

- **Database Management** (Python) – Certificate
- **Lauching Into Machine Learning (Python) –** Certificate
- **Full Stack Python (HTML, CSS, JavaScript, MySQL, Vue, Bootstrap) –** Certified
- **Full Stack .NET (HTML, CSS, JavaScript, MySQL, Bootstrap, Node.js, Handlebars) –** Certified
- **Full Stack Java (HTML, CSS, JavaScript, MySQL, Vue, Bootstrap) –** Certified
- **Spring Specialization (Java) –** Certificate
- **React JS (JavaScript) –** Certificate
- **Salesforce Platform (Apex) –** Certified
- **Unity 2D (C#) –** Certified
- **IBM Cloud (Kubernetes) –** Certified

➢ LANGUAGE

- **Certificate of Proficiency at Instrumental Level of English:** in Engineering - Certificate

➢ PERSONAL CAPABILITIES

- **Employability Skills –** Certificate
- **Soft Skills –** Certificate

➢ TECHNICAL SKILLS

- **Databases:** MySQL, SQL Server Management Studio, MongoDB, SOQL, and SOSL
- **Software Architecture:** MVC, MVVM, Layered Architecture, Microservices, API Integration
- **DevOps & Tools:** Git, GitHub
- **Cloud & Platforms**: IBM Cloud, Salesforce Platform
- **Testing:** Fundamentos de Unit Testing
- **Game Development:** Unity (C#), implementing AI logic in games
- **AI & Data:** Python, TensorFlow, fundamentos de Machine Learning, Data Analysis
- **Backend Development:** Python, Java (Spring Boot), Node.js, .NET, PHP, RESTful APIs
- **Frontend Development:** React, Vue, JavaScript, TypeScript, HTML5, CSS3, Bootstrap,
- **Responsive Design**